

FLASH FLOOD PREDICTION SYSTEM

Updated Project Report: Current Nepal Flood Situation & India Recovery Framework

Department: Computer Science & Engineering (CSE)

Institution: Swami Vivekananda Institute of Science & Technology

Project Type: Hackathon / Academic Project

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Topic: Real-time Flash Flood Prediction, Early Warning, Emergency Response and Recovery

This report expands the earlier flash-flood prediction project with current 2026 information from Nepal and an India-focused response and recovery framework. The prediction system is a student prototype and must support, not replace, warnings and directions issued by competent authorities.

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1. Introduction

A flash flood is a rapidly developing flood caused by intense rainfall, rapid runoff, river response, drainage congestion, landslides, debris flows or related hydrological conditions. The available warning time may be very short, particularly in steep mountain valleys and highly urbanised catchments.

The proposed Flash Flood Prediction System combines rainfall observations, short-range forecasts, terrain, soil wetness, drainage characteristics, water-level information and community observations. It produces a location-based risk score and communicates risk through a map/dashboard and alerts.

2. Objectives

- Collect and combine real-time and near-real-time flood-related data.
- Estimate flood risk for a location, road, neighbourhood or watershed.
- Classify risk into project-defined levels: Normal, Watch, Warning and Severe.
- Display risk on a digital map and support early-warning communication.
- Use community survey responses to understand flood experience and preparedness.
- Add an emergency-response and post-flood recovery workflow suitable for India.

3. System Methodology

Data Collection → Data Cleaning → Feature Engineering → Risk Prediction → Risk Classification → Map/Dashboard → Alert → Response/Recovery Support

Key inputs include rainfall intensity, 6-hour and 24-hour accumulation, forecast rainfall, soil moisture/saturation, elevation and slope, flow accumulation, drainage density, river/water level and local observations.

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4. Current Flash-Flood Data — India

The existing project report records the India Meteorological Department (IMD) National Flash Flood Guidance Bulletin issued on 07 September 2026 at 06:35 IST. It reported an Area of Concern (AoC) of NIL, with rainfall up to 31 mm in the preceding six hours and up to 60 mm in the preceding 24 hours over some watersheds/neighbourhoods. The bulletin was valid until 11:30 IST for that issue.

Parameter	Current project-report data
Area of Concern	NIL in the cited IMD National Flash Flood Guidance Bulletin
Rainfall — previous 6 hours	Up to 31 mm over some watersheds/neighbourhoods
Rainfall — previous 24 hours	Up to 60 mm over some watersheds/neighbourhoods
Bulletin issue	07 September 2026, 06:35 IST
Validity	Valid until 11:30 IST for that issue

The same project report notes widespread rainfall over Sub-Himalayan West Bengal & Sikkim during 6–9 September and rainfall activity over Gangetic West Bengal during 7–9 September, with isolated heavy rainfall mentioned for Gangetic West Bengal during 8–9 September. Therefore, the prototype should combine observed rainfall with forecast rainfall and local susceptibility instead of relying on a single measurement.

5. Current Nepal Flood Situation — September 2026

Nepal is currently dealing with the severe consequences of the 26 August 2026 disaster affecting the Bhotekoshi/Trishuli river systems. Recent reporting describes the event as a catastrophic flood/debris-flow disaster associated with a glacier collapse in the Langtang region. The disaster destroyed settlements and critical infrastructure and created major search-and-rescue challenges.

Situation snapshot: UNFPA's situation report covering 28 August–3 September recorded government figures of 1,204 deaths and approximately 4,216 people unaccounted for. Later reporting on 6 September reported more than 1,300 deaths and nearly 5,000 people still missing. These figures are changing as search operations continue, so the report treats them as time-stamped situation data rather than fixed final totals.

The disaster also affected homes, roads, bridges, hydropower facilities, health services and communications. Reports indicate that helicopters, excavators and temporary bridges have been used in rescue operations, while relief centres and health facilities have supported displaced people.

Why Nepal is important to the proposed prediction system

- Mountain catchments can produce extremely rapid downstream flow and debris movement.
- Flood impacts can be amplified by landslides, boulders, sediment and damaged roads/bridges.
- Hydropower tunnels and infrastructure require specialised rescue planning.
- Community-based warnings can be valuable when formal communications are delayed or incomplete.
- A useful prediction system should therefore include terrain, upstream conditions, river level and local reports—not rainfall alone.

6. Proposed Risk Model

For a prototype, a 0–100 risk score can be calculated from weighted features. A suggested structure is:

Feature	Suggested role
Rainfall intensity / accumulation	Highest influence because rapid rainfall can generate rapid runoff
Forecast rainfall	Provides lead time for warning
Soil saturation / antecedent rainfall	Indicates available infiltration capacity
Terrain, slope & flow accumulation	Identifies locations where water concentrates rapidly
Drainage density / obstruction	Captures urban waterlogging susceptibility
River / water-level trend	Detects rising downstream flood risk
Community observations	Adds local evidence such as water depth and road inundation

Project-defined classification: Normal (0–24), Watch (25–49), Warning (50–74), Severe (75–100). These are prototype categories and must be calibrated against verified local historical events; they are not official IMD warning categories.

7. If a Nepal-Type Extreme Flood Occurs in India: Response & Recovery Framework

A similar event in India—especially in the Himalayan states, the Northeast, or downstream river basins—should be managed as a multi-hazard emergency because flash flooding can occur together with landslides, debris flows, bridge failure, road blockage and infrastructure damage.

A. First 0–6 Hours: Life Safety

- Activate the local Incident Response/Emergency Operations structure and coordinate with district/state disaster-management authorities.
- Follow official IMD, Central Water Commission (CWC), NDMA and local authority alerts.
- Evacuate people from riverbanks, low-lying areas, unstable slopes and known flood channels before water rises.
- Use geo-targeted alerts, SMS/mobile notifications, sirens, local volunteers, police, radio and public-address systems.
- Stop movement across flooded roads, bridges and culverts; do not attempt to cross fast-moving water.
- Prioritise children, older persons, persons with disabilities, patients and isolated communities.
- Search and rescue should use trained teams and appropriate equipment; avoid untrained entry into unstable debris or tunnels.

B. 6–24 Hours: Emergency Stabilisation

- Establish safe shelters with drinking water, sanitation, food, first aid and essential medicines.
- Restore emergency communication through radio, satellite/mobile backup and temporary networks where needed.
- Conduct rapid damage and needs assessment for homes, roads, bridges, electricity, water supply, hospitals and schools.
- Separate safe routes from blocked or unstable routes using updated field reports and mapping.
- Monitor contaminated water, sanitation failures and disease risks in shelters and affected settlements.
- Maintain a missing-person and family-reunification system with verified records.

C. 1–7 Days: Early Recovery

- Clear priority roads and restore access to hospitals, shelters, food distribution points and isolated villages.
- Repair temporary water, sanitation, power and communication systems.
- Provide emergency cash/food/livelihood assistance to affected households.
- Inspect buildings and infrastructure before allowing reoccupation.
- Map debris-flow channels and newly unstable slopes to prevent secondary casualties.

- Provide psychosocial support and protection services for displaced and bereaved families.

D. Weeks–Months: Build Back Better

- Rebuild or relocate critical infrastructure away from high-risk flood and debris-flow corridors where feasible.
- Upgrade bridges, culverts, drainage, river monitoring and slope stabilisation based on hazard mapping.
- Use historical events, remote sensing, sensors and verified community reports to retrain the prediction model.
- Create evacuation maps, route signage, community drills and local volunteer networks.
- Integrate the project dashboard with authorised government warning channels rather than issuing unsupported public warnings.
- Conduct an after-action review and update thresholds, response plans and sensor locations.

8. India-Focused Recovery Module for the CSE Project

The proposed software can be extended beyond prediction so that the dashboard becomes a decision-support tool for response and recovery. A simple architecture is:

Sensor/Weather Data → Risk Engine → GIS Risk Map → Alert → Evacuation/Safe Route Layer → Incident Reports → Damage Assessment → Recovery Tracking

Module	Function
Live Risk Map	Shows Normal/Watch/Warning/Severe project risk zones
Alert Engine	Generates location-based notifications when calibrated thresholds are exceeded
Citizen Report	Accepts water depth, blocked road, drainage and photo/location reports
Safe Route	Displays designated evacuation/safe locations supplied by authorities
Damage Report	Records damaged homes, roads, bridges, power and water infrastructure
Recovery Tracker	Tracks restoration status of roads, utilities, shelters and services
Analytics	Stores historical events and survey responses for model validation

9. Community Survey

Survey objective: collect local information on previous flooding, warning awareness, preparedness, drainage problems and the expected usefulness of a flash-flood prediction application.

Survey creation link: Google Forms — <https://forms.google.com/> (create and publish the form, then replace the placeholder with the generated respondent link).

1. Age group: Under 18 / 18–25 / 26–40 / 41–60 / Above 60
2. Area/locality and district: _____
3. Have you experienced flooding or waterlogging? Yes / No
4. Frequency: Rarely / Sometimes / Frequently / Every monsoon
5. Typical maximum water depth: <15 cm / 15–30 cm / 30–60 cm / >60 cm / Not sure
6. Main perceived cause: Heavy rainfall / Poor drainage / River overflow / Encroachment / Other
7. Have you received a flood warning before an event? Yes / No / Not sure
8. Preferred warning channel: SMS / Mobile app / Siren / TV-radio / Social media / Local authority
9. Useful warning lead time: <30 min / 30–60 min / 1–3 h / >3 h
10. Do you know the safest evacuation route? Yes / No / Not sure
11. Do you keep an emergency kit? Yes / No / Partly
12. Confidence in early flash-flood prediction: 1 / 2 / 3 / 4 / 5
13. Would you use a flash-flood risk map on mobile/web? Yes / No / Maybe
14. Most useful feature: Risk map / Rainfall forecast / Alerts / Safe routes / Emergency contacts / All
15. Local flood problem or suggestion: _____

10. Expected Output of the CSE Project

- Live dashboard with map-based flood-risk zones.
- Risk score with an explanation of major contributing factors.
- Rainfall history and short-range forecast charts.
- User reporting module for local waterlogging/flood observations.
- Alert generation when calibrated project thresholds are exceeded.
- Historical database for rainfall, water levels, flood events and survey responses.
- Emergency-response layer containing official evacuation routes and shelter information.
- Recovery dashboard for infrastructure damage and restoration status.
- Future machine-learning module trained only on verified historical flood-event labels.

11. Limitations and Safety

- A student prototype cannot guarantee that a flood will or will not occur.
- Forecast uncertainty, missing sensors and inaccurate citizen reports can affect predictions.
- Risk thresholds must be validated using local historical observations.
- The Nepal situation figures are time-sensitive and may change as search and recovery operations continue.
- Emergency actions must follow official instructions from IMD, CWC, NDMA, state/district disaster-management authorities and local emergency services.
- The system should not generate an evacuation order independently of authorised authorities.

12. Conclusion

The Flash Flood Prediction System provides a practical CSE/hackathon framework for combining meteorological, hydrological, terrain and community data. The current India project data show why rainfall observations and forecasts must be interpreted with local susceptibility. The catastrophic Nepal disaster demonstrates the additional importance of upstream monitoring, terrain/debris-flow risk, resilient communications and rapid search-and-rescue planning. If a comparable event occurs in India, recovery should progress from life safety and emergency shelter to restoration of access and essential services, followed by resilient reconstruction and model improvement. The proposed dashboard can support this complete cycle while remaining subordinate to official warnings.

13. References / Current Sources

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